

An Improved Best Path Methodology for Mobile GIS Based On A* Algorithm

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Abstract—An improved best path method for mobile geographic information system (GIS) based on A* algorithm is studied in this paper. Key technologies are presented in the methodology, including mobile GIS framework, spatial data bases, electronic map display, global positional system (GPS) communications, and the best path based on A* algorithm. In order to deal with the problem between the limited processing speed and low memory of pocket pc with the high speed inquire of mobile GIS, an improved best path method based on A* algorithm is further proposed that has good stability, practicality and efficiency, and thus suitable for pocket pc. Considering of the the hardware limited property of pocket pc, the best path and other related method are built originally from the base. A mobile GIS system integrated with the improved A* algorithm is therefore developed authorship by embedded visual c++ . A field test is carried out through this system by Jiaozuo spatial data, and the results show that the proposed methodologies are feasible and adaptable for mobile GIS.

Keywords- mobile GIS, Heuristic Search Algorithm, A* Algorithm, GPS, Access DataBase

I. INTRODUCTION

During the information era with increasing developing speed, how to get the information undoubtedly becomes a hot issue [1]. People want to get spatial information more and more under mobile environment. The traditional GIS begin to change toward to mobile GIS [2]. The typical use and mature business of mobile GIS field development is the field data collection, vehicle navigation and positioning system integration applications [3].

With the development of information, The use of mobile GIS will be more and more widely [4]. The network analysis and transportation analysis is becoming an essential application. The selection of the shortest path is an important part in the network analysis and transportation analysis. Because of the limitations of PDA hardware, how to improve the operating efficiency is a key problem [5].

In order to develop a visualization of spatial information systems based on embedded products such as personal digital assistant (PDA), the mobile computing technology combined with GIS electronic map will built out the basic framework of GIS system. It bring out the visualization of spatial information system based on embedded products such PDA. It has also been focused on how to save time best, more

efficient, so that people keep an eye on the choice of best path [6]. Integrated with more specific functions, the mobile GIS system implied in this paper can be used in many industries, transportation, public security, fire, geology, mapping, communications and many other industries [7].

Considering of the hardware of pocket pc, the problem of shortest path selection has begun to shift from the desktop GIS to the mobile GIS. For the shortest path, most existing studies using dijkstra algorithm and its improvement. This paper is focused on the best path method based on A* algorithm under mobile gis environment. The basic framework of mobile GIS, spatial databases, electronic map display, GPS communications and other related technologies are further presented. A field test will be carried out through the mobile GIS sytem by Jiaozuo spatial data.

II. MOBILE GIS SYSTEM ARCHITECTURE BASED ON A* ALGORITHM

Compared with the desktop GIS, mobile GIS has the properties which is the mobility, diversity of clients, services, real-time, distributed data resources, diversity, diversity of information carriers. The mobile GIS functionality is simple. The mobile gis system architure is shown in figure 1.

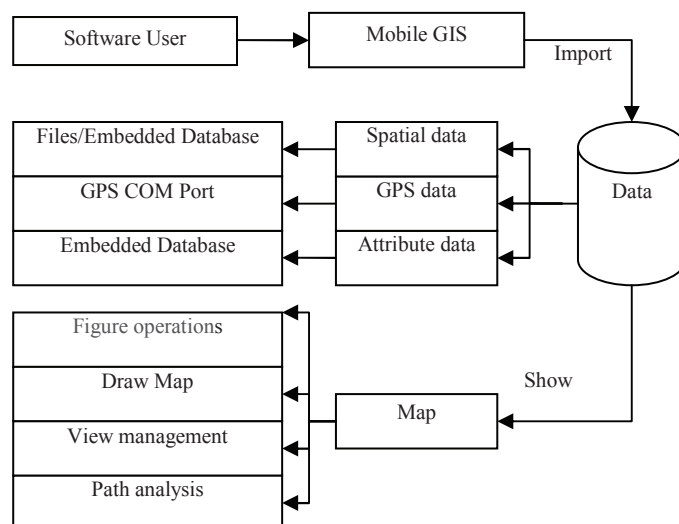


Fig.1 Logical structure

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Mobile GIS run on various mobile terminal which has limited computing power, smaller screen and small storage. The amount of spatial data is small and portable. The function modules of the mobile GIS system is shown in figure 2.

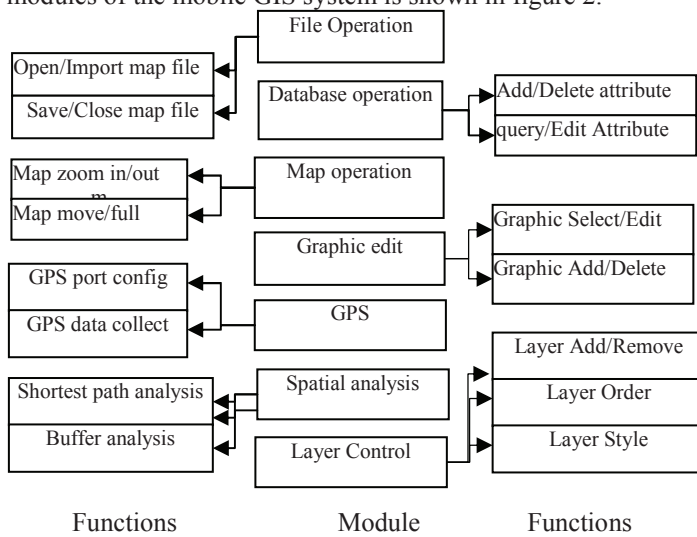


Fig.2 Functional structure

III. THE KEY TECHNOLOGY OF MOBILE GIS SYSTEM BASED ON A * ALGORITHM

A. Spatial Data Storage

The mobile GIS system developed by the authors is integrated with the embedded database technology. Firstly, the system is debugged in the VC++6.0 environment. Secondly, the system will be replanted into embedded VC++4.0. Some corresponding codes would be modified in the replanting process. The implementation steps are as follows:

1).DLLin~setup\DataAccess3.1\ProgramFiles\DataAccess 31\DEVICE\ARM\SA1100\CE\RETAIL of adocepzip.exe or adocedkzip.exe installation package's added to PDA's windows directory.

2).adoce31.h, adocec31.h and all the files in vcado added to the project. adoce30.h changed to Adoce31.h. The third sentence in vorecordset.h will be changed as this,
 BOOL CVORecordset::g_Init= FALSE;
 CLSID CVORecordset::g_ClsID; TCHAR*
 CVORecordset::g_ProgID=TEXT("ADOCE.Recordset.3.1");

3).Access database imported into PDA, transformation is taking. In order to get .mdb form database, regsvrce.exe, database and executable file should be in the same directory.

4). The connection of database is as follow:

```

CVOConnection*m_Conn;
CVORecordset *m_Set;
CString cdbName; cdbName=_T("wf.cdb");
m_Conn=new CVOConnection(cdbName);
m_Set=new CVORecordset(*m_Conn);
m_Set->Initialize();
m_Set->Open(s_Sql, adOpenForwardOnly, adLockPessimist
  
```

B. Spatial Data Organization

The so-called map, the most popular explanation is abstracting the reality of things, and translating objects to space

objects or spatial entities, such as housing, grass, roads and so on. These meaningful point, line and area entities can be represented by using map features, such as point, line, area.

According to object-oriented programming idea, the open geographic consort (OGC) standard is adopted. The inheritance, encapsulation and polymorphism of object-oriented programming are adopted into design the base class CMapObject for all graphical elements. The CMapPoint (points), CMapLayer (layer), CmapPLine (broken line), CMapRegion (polygon), CMapText (fonts), and so on. The node list, point, line, surface structure is therefore defined, as shown in figure 3.

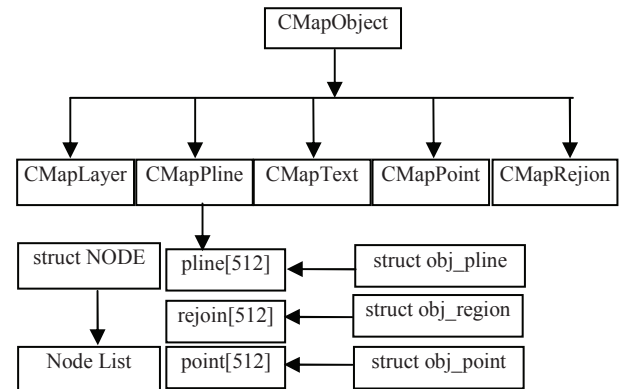


Fig 3 feature structure frame

C. Obtain the Shortest Path

A* algorithm is firstly proposed by Hart, Nilsson, Raphael. It is different from the blind-type search method, such as dijkstra algorithm. It is not a breadth-first search, but is a heuristic search algorithm widely used in shortest path. It is used widely in AI and game.

The basic idea is: using the appropriate evaluation function and good design. The actual price from the initial node to node n and the estimated cost from n node to the destination node can be got [6].

In order to get best path from the starting point to the specified destination, the evaluation function is expressed as: $f(j) = g(j) + h(j)$. Its meaning is as follows: (1) g means the distance of the initial node to temporary node (2) h means the distance between Interim point and the target node (an estimated value) (3) f means the total assessed value.

Selecting the appropriate evaluation function is directly related to the success of A* algorithm. How to determine the function is according to the actual situation. Thus, it is very important to select the heuristic function. An inappropriate heuristic does not play a catalytic role to A* algorithm. It will slow down the A* algorithm [7].

If the perfect path is desired, the heuristic function value should be a minimum valuation from a temporary location to the destination node. It is less than actual costs, that is $h^*(n) < H(N)$.

If the heuristic estimates are accurate, then A* will not deviate from the optimal path. It's difficult to get an accurate heuristic valuation. If the right message is given to A*, the operation will be correctly.

If the value of heuristic estimates tend to be accurate, the A* search operation will be less. The efficiency will be higher.

The f value increasing is related with the number of search. When the heuristic is accurate, then f will not change. When the heuristic value is much lower than the actual value, f will be rapidly increased.

A * algorithm implementation steps are as follows,

The NODE structure and object OPEN and CLOSE table are defined in the map file which store the node, defined obj_point, obj_pline, obj_region structure. The object point, pline, region are defined to store point, line and plane. For example, the points of pline will be stored to the pline array, the distance of start point to each point can be calculated in the road to determine the best path selection.

1). Firstly, the first point on the road layer is selected as a starting point, and is deposited in OPEN statement, the cost is 0, that is, g is 0. Order CLOSED table is empty, set the cost of other node'∞.

2) The distance g from the starting point to various points on the road (these points stored in the array pline) is calculated. The distance from point to the destination node h can be estimated by $f(j) = g(j) + h(j)$. Thus the f valude is got. When the OPEN table is empty. There is no path from the starting point to the target point. Otherwise, the selected points with minimum f value of OPEN table, called BEST. If the point is the target node, then go to step 3), whereas the according to connection property recorded in the database to find the successor node BEST. For each successor node n, as is the following process:

a) The cost of computing nodes n, the formula is $g(n) = \text{BEST cost} + \text{the cost from the BEST to n}$.

b) If n and any node of the same is in table OPEN, then whether n has the lowest g value is determined. If n nodes is in the same cost of OPEN table, and the establishment matching node back pointer for BEST.

c) If the n match with a node in the table CLOSED, then whether n has the lowest g value determined. If the same node n is instead of costs, the matching node back pointer to BEST. The matching node shift to the table OPEN.

d) If n is not in OPEN and CLOSED, the backward pointer of n point will be put to BEST, and n will be put into the OPEN table. The evaluation function $f(n) = g(n) + h(n)$ of node n is calculated. The second step will be repeated.

3) The BEST nodes will be back. The nodes from backward pointer of the BEST to the original nodes are traversed, which are to be the best path.

The following state space is as shown in figure 4, (the starting position is A, the target location is P, the number of letters, setting after the assessed value of the node).

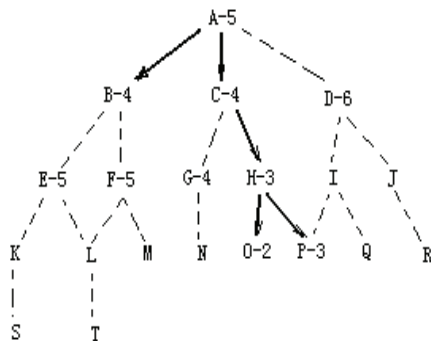


Fig.4 State space diagram

- (1) The initial situation OPEN = [A5] CLOSED is empty
- (2) Searching all the child nodes of A5 is the B4, C4, D6 and putting them in OPEN statement, the A5 will be put in CLOSED table
- (3) According to such a step, followed by estimation of B4, C4, H3, O2, the node will be put in CLOSED into the table, the final estimate P3, the final solution

The relevant Main achievement of code is as follow,

```
struct NODE { //Node List
double f,h;
double g,tmpg;
double x,y; struct NODE *Parent;
struct NODE *NextNode; /* for filing purposes */ ;
struct obj_pline
{CString name;
int pointnum;
int penwidth;
int pencolor;
double *longitude;
double *latitude;
obj_point * next; };

for(int n=0;n<m_pDoc->ONum;n++)
{ int p=pline[n].pointnum; // the distance
between the clicking point to other points are
stored in the array
for(int i=0;i<p;i++) // calculate point to all the
nodes from the line in Gaphic
{f1=*(pline[n].longitude+i);
f2=*(pline[n].latitude+i);
dd[n][i]=m_pDoc->GetDistance(jing,wei,f1,f2)*100000;
//Put data into an array of characters to determine the
nearest point }}
min=dd[0][0]; //The first value is assigned to the
smallest variable in order to compare in the following
for(int m=0;m<m_pDoc->ONum;m++) // Find out the
minimum in array
{ int p=pline[m].pointnum; for(int i=0;i<p;i++)
{if(dd[m][i]<min) {min=dd[m][i]; }}}
```

IV. THE IMPLEMENTATION MOBILE GIS SYSTEM BASED ON A *ALGORITHM

According to the improved best path method based on A* algorithm, integrated with GPS and other key technologies, a mobile GIS system based on pocket pc is developed in the embedded visual c++ enviroment. Some spatial data of one certain region of Jiaozuo are adopted into the mobile GIS system. Figure 5 is the showing of map, figure 6 is the map querying, figure 7 is the menu of best path analysis, figure 8 is the result of the best path analysing.

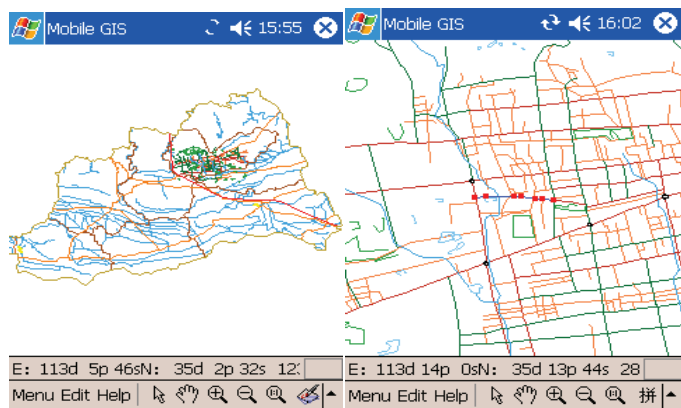


Fig.5 The shown of Map

Fig.6 Map query

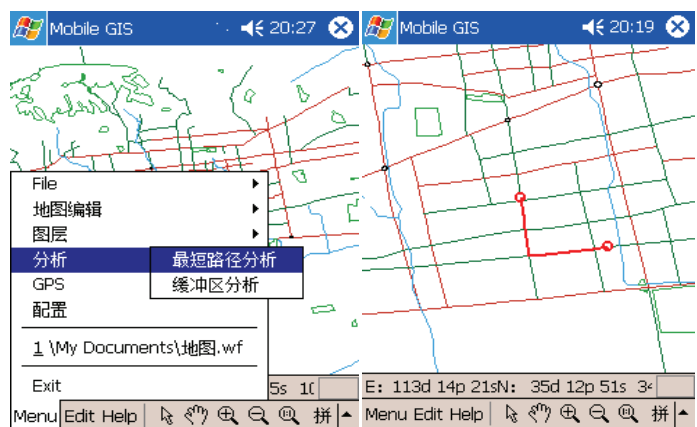


Fig.7 Menu of shortest path analysis

Fig.10 The Result of Path Search

V. CONCLUSION

As an important part of GIS, mobile GIS can supply spatial information service and application under mobile computing environment, and thus brings convenience to GIS user. This paper proposed an improved best path method based on A* algorithm for mobile GIS. Several key technologies are researched in the method, including mobile GIS framework, spatial data bases, electronic map display, global positional system (GPS) communications, and the best path based on A* algorithm. Considering of the the hardware limited property of pocket pc, the best path and other related method are built originally from the base.A mobile GIS system integrated with the improved A* algorithm is therefore developed authorship by embedded visual c++ . A field test is carried out through this system by Jiaozuo spatial data, and the results show that the proposed methodologies are feasible and adaptable for mobile GIS.

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